

Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)
Lecture 8: Equilibrium of
Rigid Body in 3D
By Saeed Lotfan

Equilibrium in 3D: Free-Body Diagrams

The first step in solving three-dimensional equilibrium problems, as in the case of two dimensions, is to draw a free-body diagram. Before we can do this, however, it is first necessary to discuss the types of reactions that can occur at the supports.

Support Reactions

- A support **prevents** the translation of a body by exerting a force on the body.
- A support **prevents** the rotation of a body by exerting a couple moment on the body.

Equilibrium in 3D: Free-Body Diagrams

Support Reactions

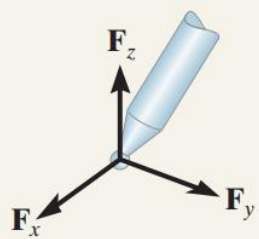
TABLE 5–2 Supports for Rigid Bodies Subjected to Three-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(1)  cable		One unknown. The reaction is a force which acts away from the member in the known direction of the cable.
(2)  smooth surface support		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.

Equilibrium in 3D: Free-Body Diagrams

Support Reactions

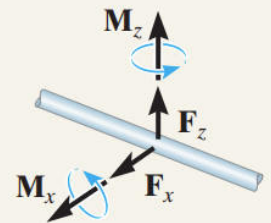
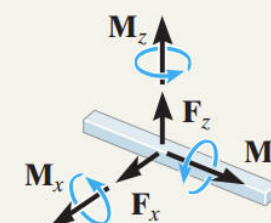
TABLE 5–2 Supports for Rigid Bodies Subjected to Three-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(3)  roller		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(4)  ball and socket		Three unknowns. The reactions are three rectangular force components.

Equilibrium in 3D: Free-Body Diagrams

Support Reactions

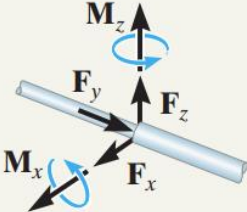
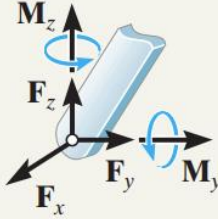
TABLE 5–2 Supports for Rigid Bodies Subjected to Three-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(5)  single journal bearing		Four unknowns. The reactions are two force components, because a journal bearing allows movement along the axis of the shaft, and two couple-moment components which act perpendicular to the shaft. <i>Note:</i> The couple moments are <i>generally not applied</i> if the body is supported elsewhere. See the examples.
(6)  single journal bearing with square shaft		Five unknowns. The reactions are two force and three couple-moment components. <i>Note:</i> The couple moments are <i>generally not applied</i> if the body is supported elsewhere. See the examples.

Equilibrium in 3D: Free-Body Diagrams

Support Reactions

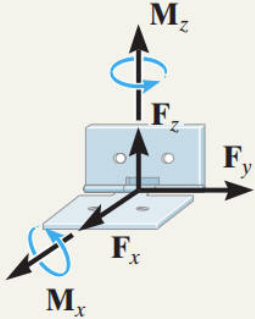
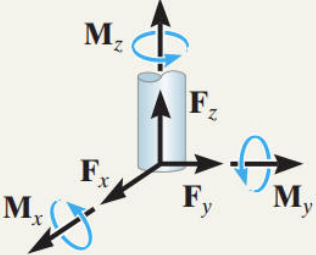
TABLE 5–2 Supports for Rigid Bodies Subjected to Three-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(7)  single thrust bearing		Five unknowns. The reactions are three force components, because a thrust bearing prevents movement along the axis of the shaft, and two couple-moment components. <i>Note:</i> The couple moments <i>are generally not applied</i> if the body is supported elsewhere. See the examples.
(8)  single smooth pin		Five unknowns. The reactions are three force and two couple-moment components. <i>Note:</i> The couple moments <i>are generally not applied</i> if the body is supported elsewhere. See the examples.

Equilibrium in 3D: Free-Body Diagrams

Support Reactions

TABLE 5–2 Supports for Rigid Bodies Subjected to Three-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(9)  single hinge		Five unknowns. The reactions are three force and two couple-moment components. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.
(10)  fixed support		Six unknowns. The reactions are three force and three couple-moment components.

Equilibrium in 3D: Free-Body Diagrams

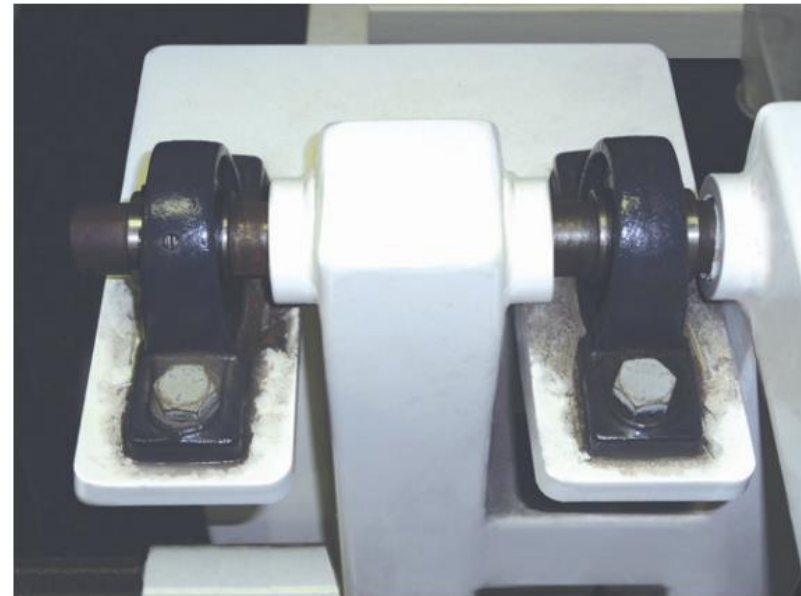
Important Note:

The single bearing supports in items (5) through (7), the single pin (8), and the single hinge (9) are shown to resist both force and couple-moment components. If, however, these supports are used with other bearings, pins, or hinges to hold a rigid body in equilibrium and these supports are properly aligned when connected to the body, then the force reactions at these supports alone are adequate for supporting the body. In other words, the couple moments will not develop since the body is prevented from rotating by the other supports.

Equilibrium in 3D: Free-Body Diagrams



This ball-and-socket joint provides a connection for the housing of an earth grader to its frame. (4)



These journal bearings are used to support the drive shaft on a machine. (5)

Equilibrium in 3D: Free-Body Diagrams

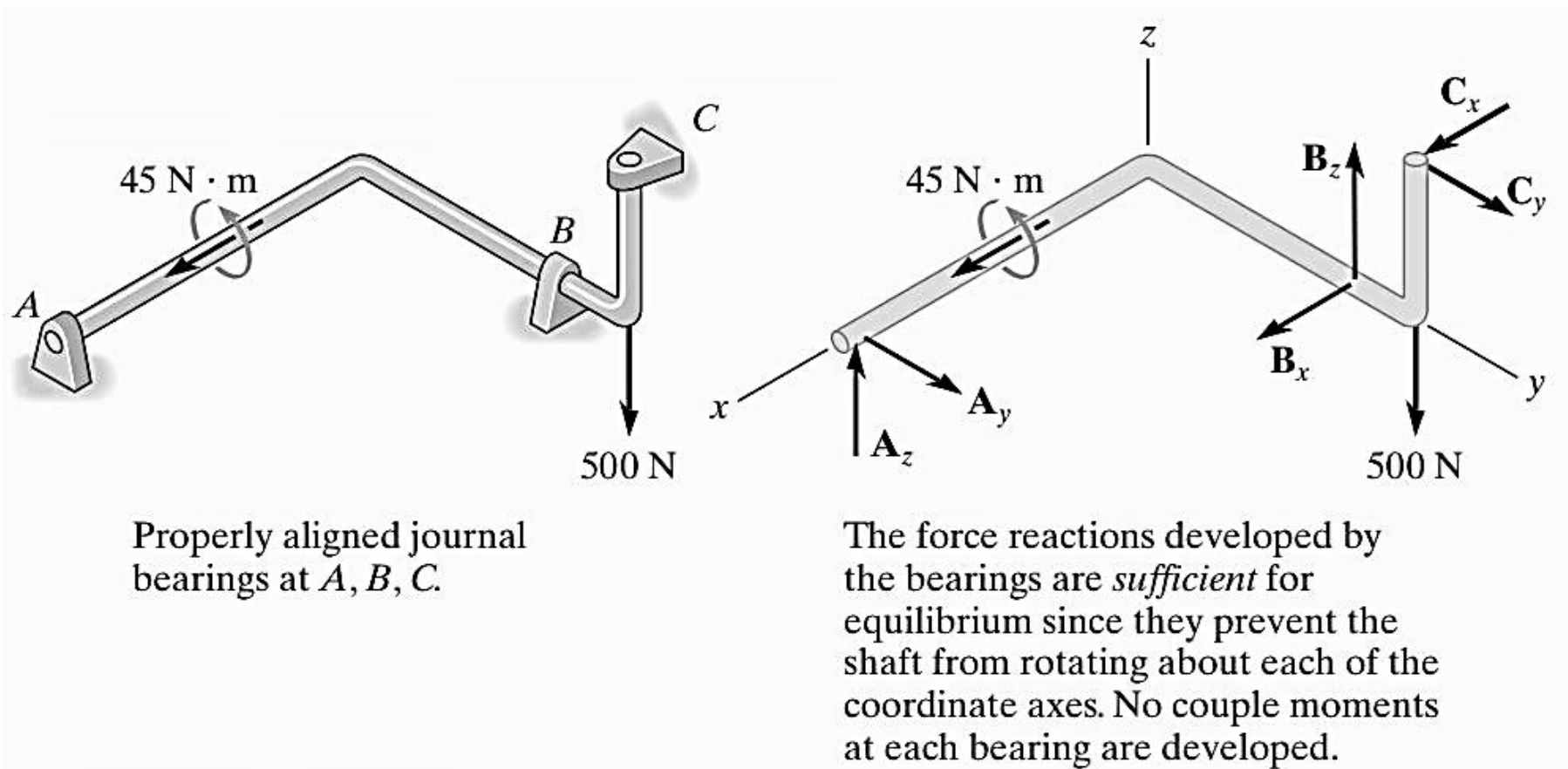


The thrust bearing supports the ends of the shaft. (7)

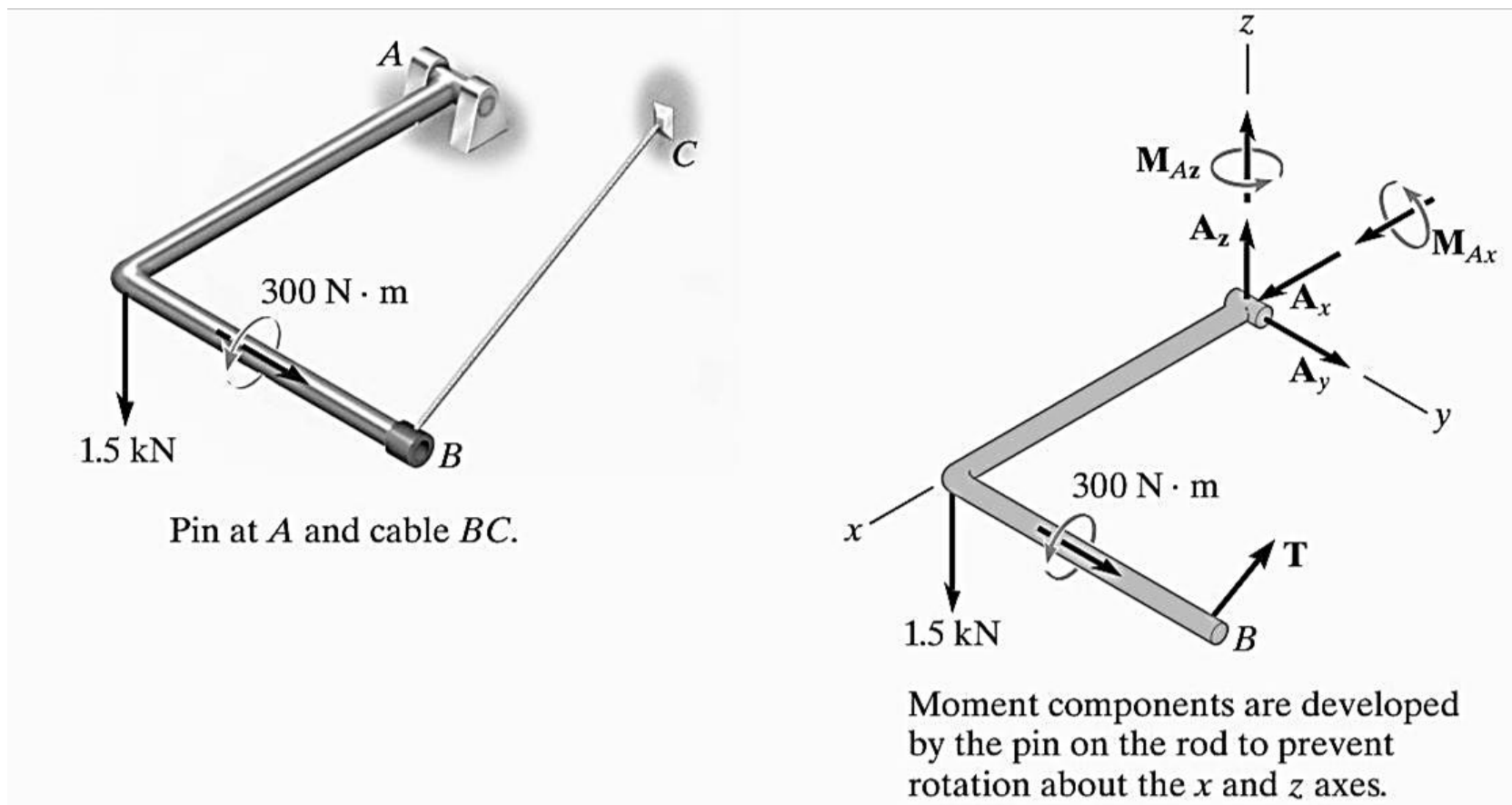


This pin is used to support the end of the strut used on a tractor. (8)

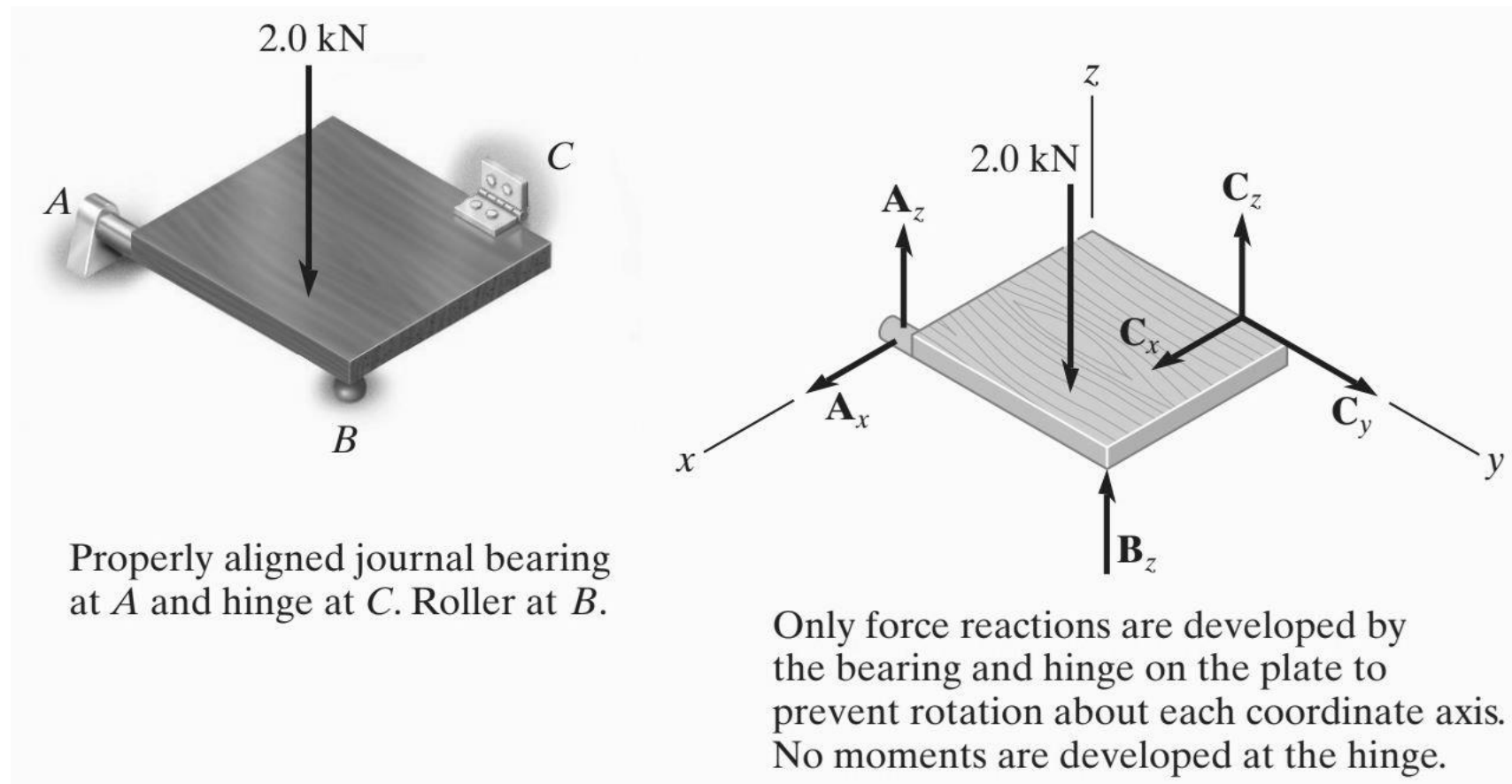
Example: Consider the rod, along with its associated free-body diagram, shown below. The x , y , z axes are established on the diagram and the unknown reaction components are indicated in the *positive* sense. The weight is neglected.



Example: Consider the rod, along with its associated free-body diagram, shown below. The x , y , z axes are established on the diagram and the unknown reaction components are indicated in the *positive* sense. The weight is neglected.



Example: Consider the plate, along with its associated free-body diagram, shown below. The x , y , z axes are established on the diagram and the unknown reaction components are indicated in the *positive* sense. The weight is neglected.



Equilibrium in 2D: Equations of Equilibrium

Vector Equations of Equilibrium

$$\Sigma \mathbf{F} = \mathbf{0}$$

$$\Sigma \mathbf{M}_O = \mathbf{0}$$

Scalar Equations of Equilibrium

$$\Sigma F_x = 0$$

$$\Sigma M_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma M_y = 0$$

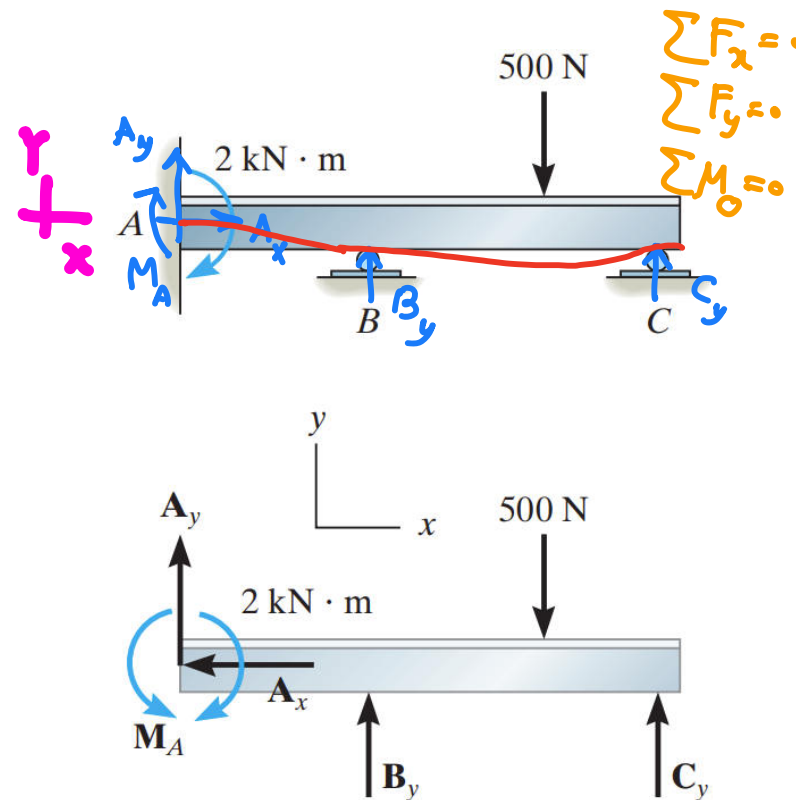
$$\Sigma F_z = 0$$

$$\Sigma M_z = 0$$

Constraints and Statical Determinacy

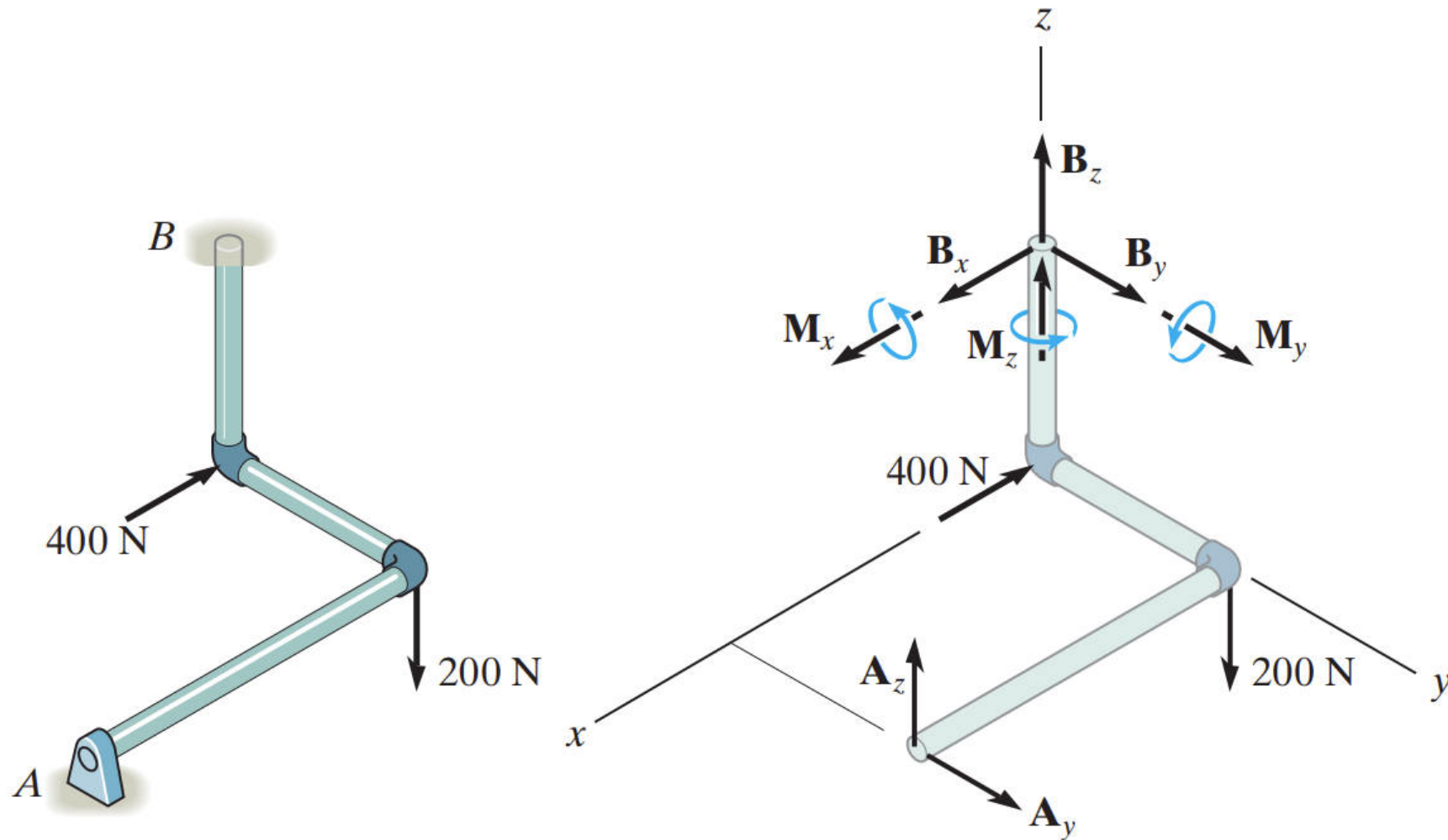
Redundant Constraints

When a body has redundant (jobless) supports, that is, more supports than are necessary to hold it in equilibrium, it becomes ***statically indeterminate***.



Constraints and Statical Determinacy

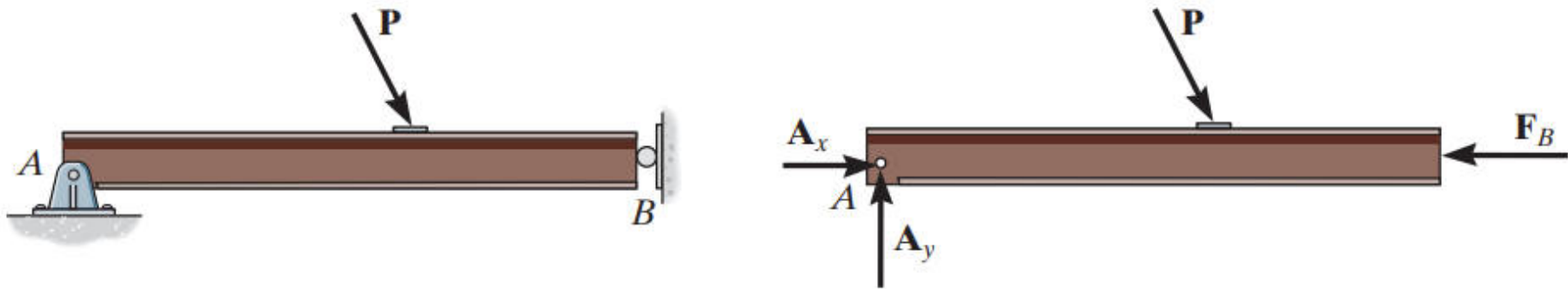
Redundant Constraints



Constraints and Statical Determinacy

Improper Constraints

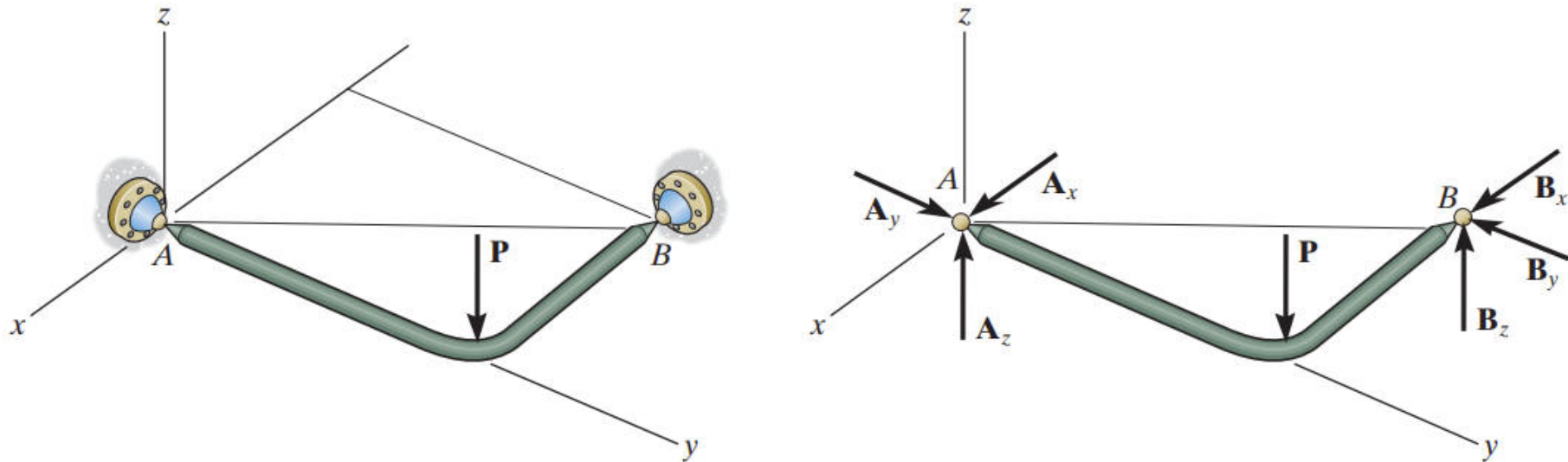
Having the same number of unknown reactive forces as available equations of equilibrium does not always guarantee that a body will be stable when subjected to a particular loading.



the pin support at A and the roller support at B for the beam are placed in such a way that the lines of action of the reactive forces are *concurrent* at point A .

Constraints and Statical Determinacy

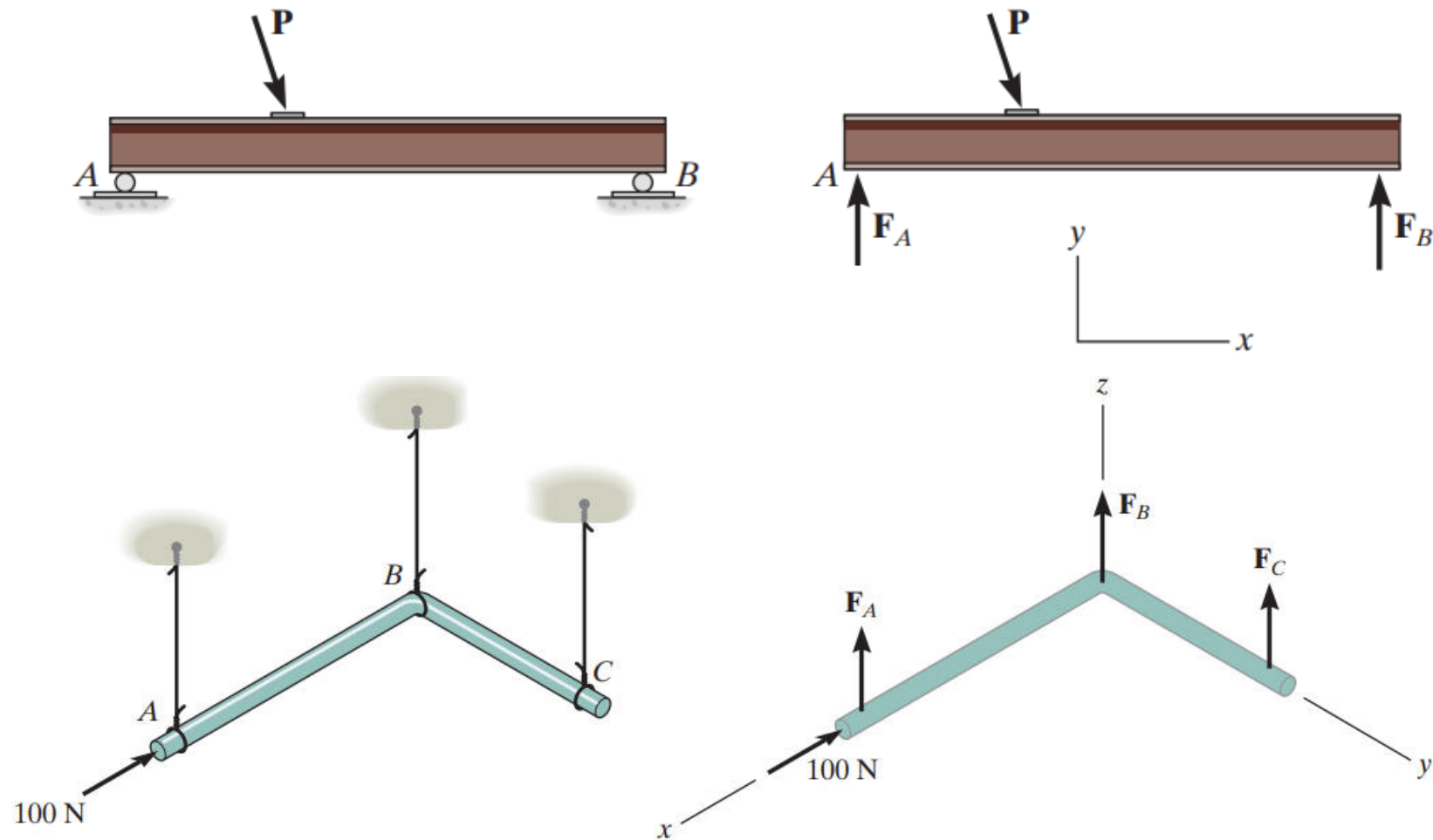
Improper Constraints



In three dimensions, a body will be improperly constrained if the lines of action of all the reactive forces intersect a common axis.

Constraints and Statical Determinacy

Improper Constraints



Concluding Notes

- Always draw the free-body diagram first when solving any equilibrium problem.
- If a support *prevents translation* of a body, then the support exerts a *force* on the body.
- If a support *prevents rotation*, then the support exerts a *couple moment* on the body.
- If a body is subjected to more unknown reactions than available equations of equilibrium, then the problem is *statically indeterminate*.
- A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.

Example: The homogeneous plate shown below has a mass of 100 kg and is subjected to a force and couple moment along its edges. If it is supported in the horizontal plane by a roller at A , a ball-and-socket joint at B , and a cord at C , determine the components of reaction at these supports.

$$\sum F_x = B_x = 0$$

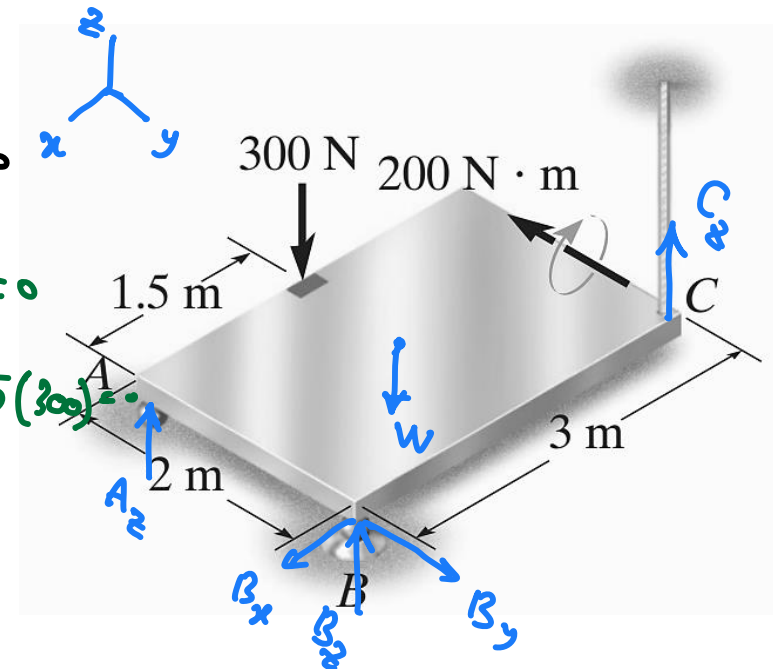
$$\sum F_y = B_y = 0$$

$$\sum F_z = A_z + B_z + C_z - 300 - W = 0$$

$$\sum M_{B_x} = -2A_z + 1W + 2(300) = 0$$

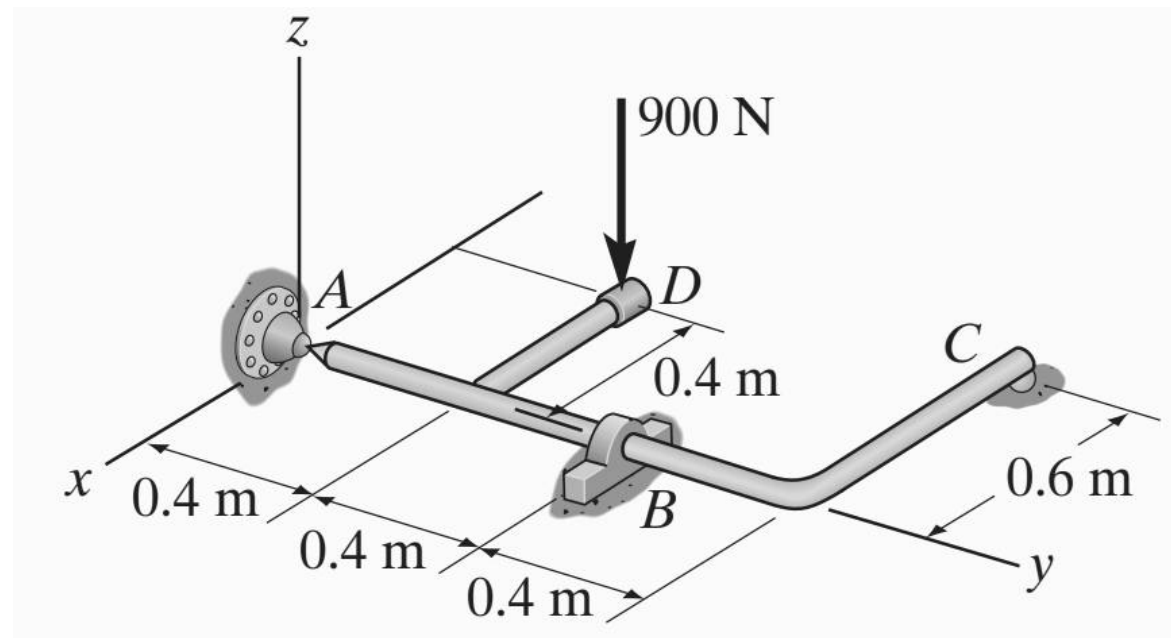
$$\sum M_{B_y} = 3C_z - 1.5W - 200 - 1.5(300) = 0$$

$$\sum M_{B_z} = 0$$



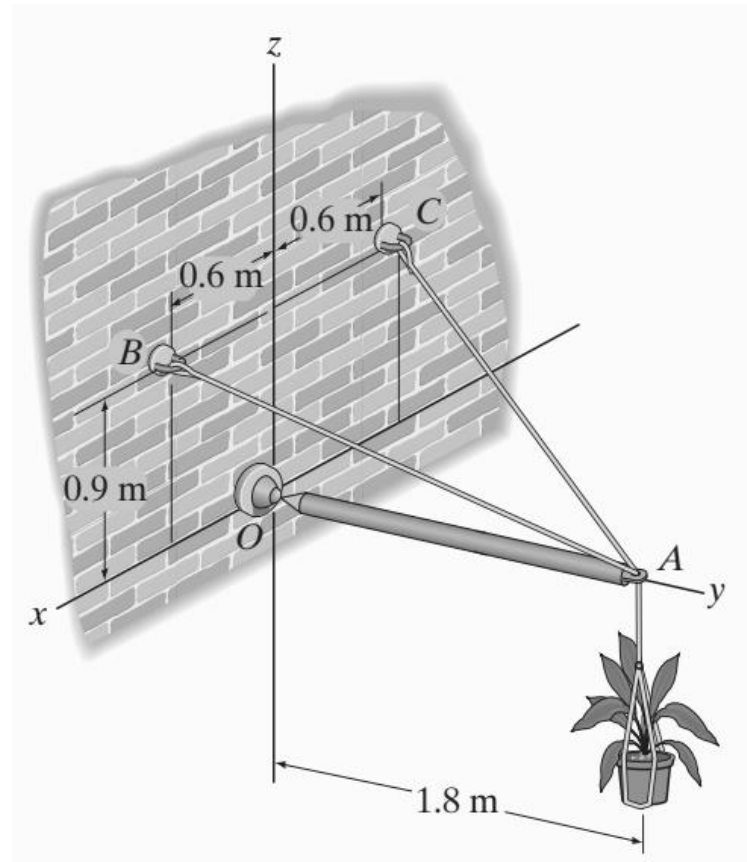
Practice at Home

1. Determine the components of reaction that the ball-and-socket joint at A , the smooth journal bearing at B , and the roller support at C exert on the rod assembly shown below.



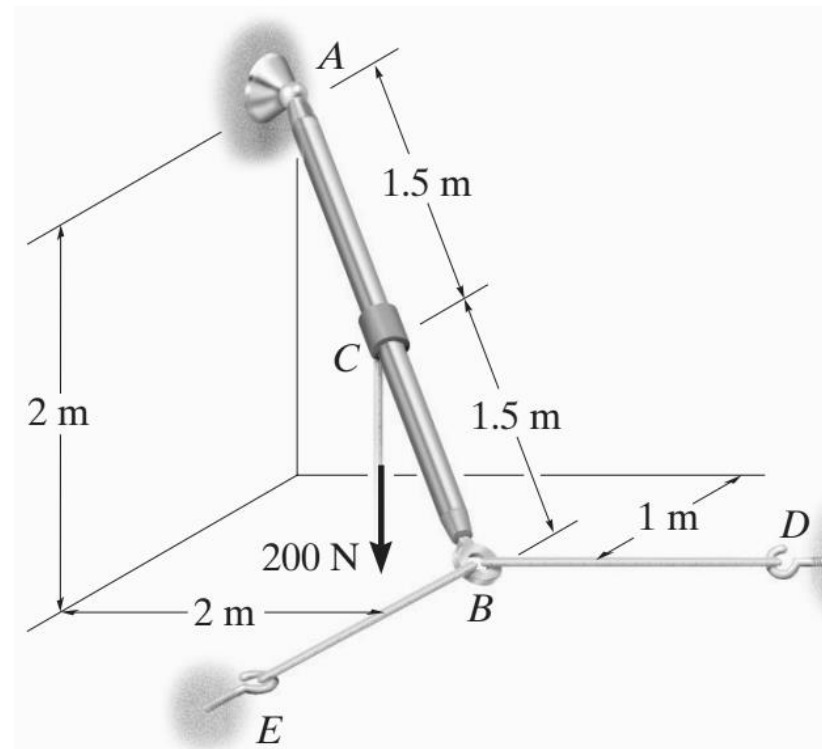
Practice at Home

2. The boom is used to support the 40-kg flowerpot shown below. Determine the tension developed in wires AB and AC .



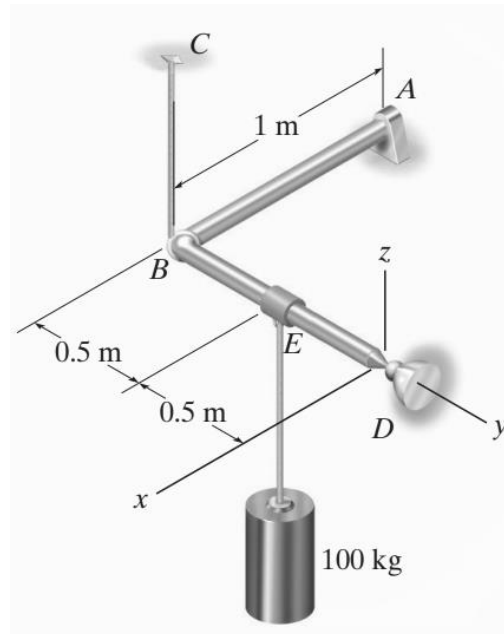
Practice at Home

3. Rod AB shown below is subjected to the 200-N force. Determine the reactions at the ball-and-socket joint A and the tension in the cables BD and BE . The collar at C is fixed to the rod.



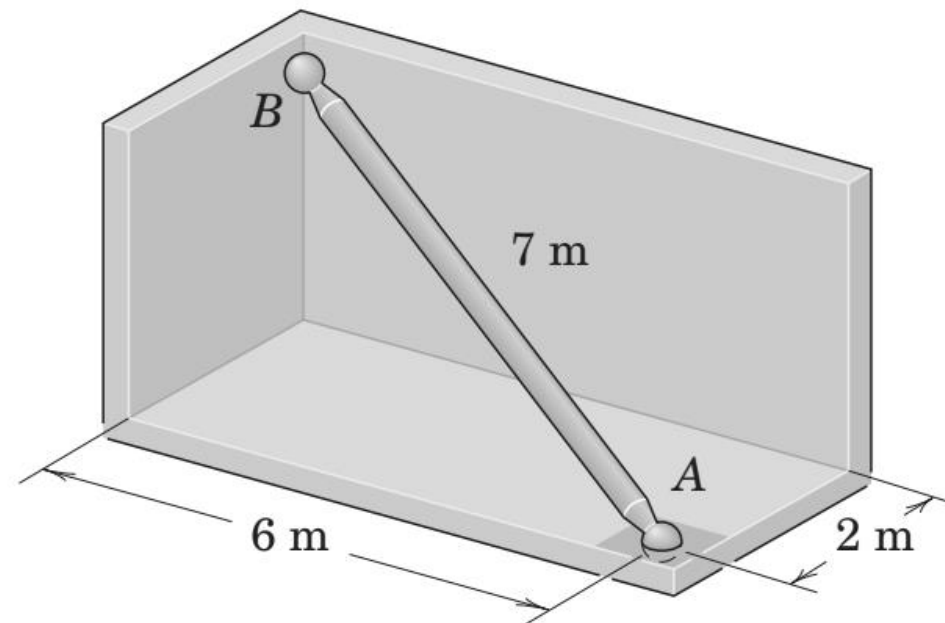
Practice at Home

4. The bent rod shown below is supported at A by a journal bearing, at D by a ball-and-socket joint, and at B by a cable BC . Using only *one equilibrium equation*, obtain a direct solution for the tension in cable BC . The bearing at A is capable of exerting force components only in the z and y directions since it is properly aligned on the shaft. In other words, no couple moments are required at this support.



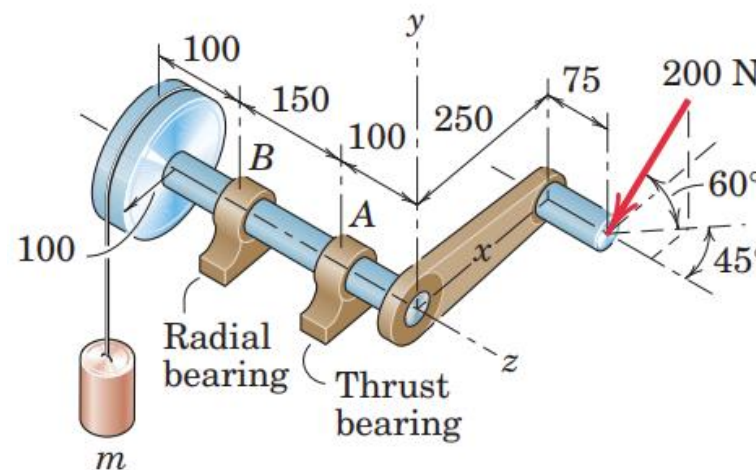
Practice at Home

5. The uniform 7-m steel shaft has a mass of 200 kg and is supported by a ball-and-socket joint at A in the horizontal floor. The ball end B rests against the smooth vertical walls as shown. Compute the forces exerted by the walls and the floor on the ends of the shaft.



Practice at Home

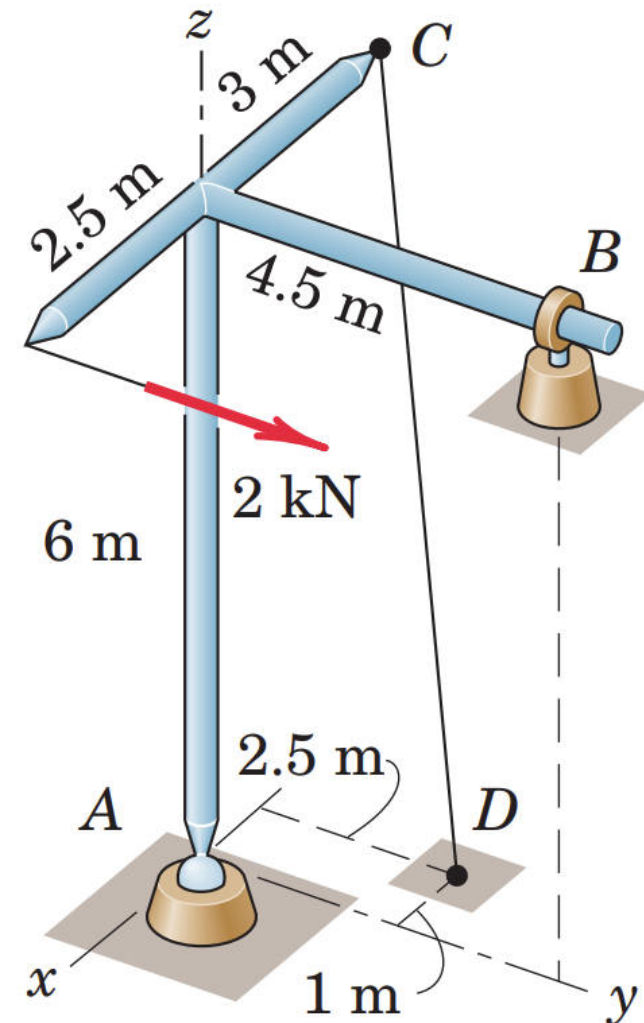
6. A 200-N force is applied to the handle of the hoist in the direction shown. The bearing A supports the thrust (force in the direction of the shaft axis), while bearing B supports only radial load (load normal to the shaft axis). Determine the mass m which can be supported, and the total radial force exerted on the shaft by each bearing. Assume neither bearing to be capable of supporting a moment about a line normal to the shaft axis.



Dimensions in millimeters

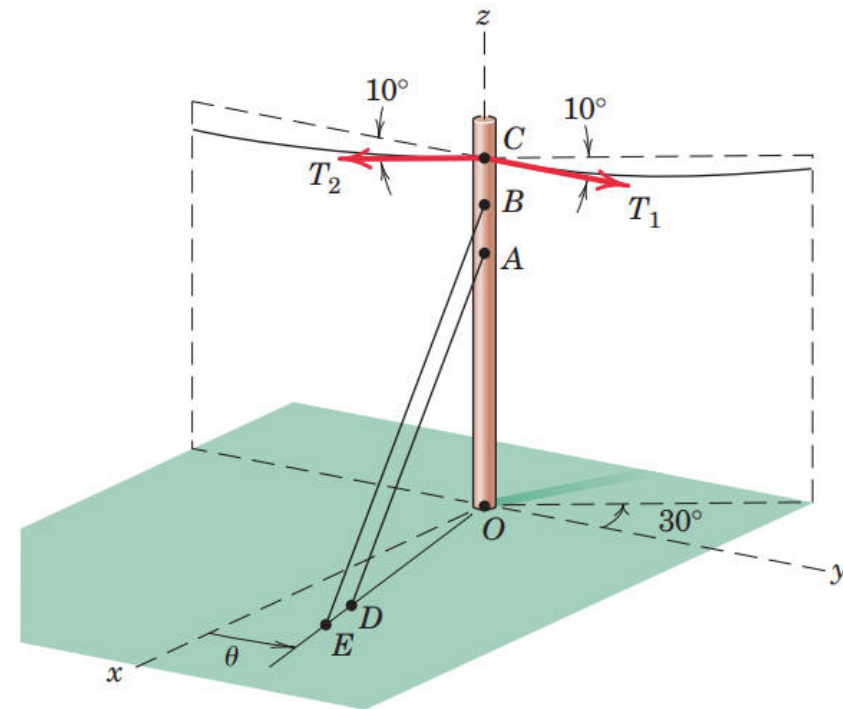
Practice at Home

7. The welded tubular frame is secured to the horizontal x - y plane by a ball-and-socket joint at A and receives support from the loose-fitting ring at B. Under the action of the 2-kN load, rotation about a line from A to B is prevented by the cable CD, and the frame is stable in the position shown. Neglect the weight of the frame compared with the applied load and determine the tension T in the cable, the reaction at the ring, and the reaction components at A.



Practice at Home

8. The vertical pole, utility cable, and two guy wires are shown here. As part of a design study, the following conditions are considered. The tension T_2 is a constant 1000 N, and its 10° angle is fixed. The 10° angle for T_1 is also fixed, but the magnitude of T_1 is allowed to vary from 0 to 2000 N. For each value of T_1 determine and plot the magnitude of the equal tensions T in cables AD and BE and the angle θ for which the moment at O will be zero. State the values of T and θ for $T_1 = 1000$ N.



$$\begin{array}{ll} \overline{OA} = 9 \text{ m} & \overline{OD} = 8 \text{ m} \\ \overline{OB} = 11 \text{ m} & \overline{OE} = 10 \text{ m} \\ \overline{OC} = 13 \text{ m} & \end{array}$$